PH4209 Assignment — 04
22MS037

Problem 1 — Moran Process with selection and Mutation

Take a population of N=10,000 individuals all of which consist of type 0 initially. For simplicity, assume that the sequence length of all individuals is L=1. Assume the mutation rate from type 1 to type 0 is 0. Let $u = 0.01$ be the mutation rate from type 0 to type 1 and $f_0 = 1.001$ is the fitness of type 0 and $f_1 = 1$ is the fitness of type 1.

1. Write a program to obtain the time-evolution of the frequencies of the two types in the population subject to both mutation and selection. Run the simulation for as long as it takes for frequencies to equilibrate.
2. Repeat the above simulation for $u = 0.01$ and $f_0 = 1.1$. Assume that half of the initial population are type 0 and the remaining half are type 1.
3. Repeat the simulation 2. with $N = 100$.

In all cases, plot the evolution of frequency of type 0 and type 1 with time. Compare your results for the equilibrium frequency in either case with the theoretical predictions obtained from analysing the quasi-species equation!

We shall simulate evolution with both selection and mutation using the Wright-Fisher process. The quasi species equation is given by,

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = \sum_j f_j u_{ji} x_j - \varphi x_j \tag{1}$$

where , x_j is the frequency of type j in the population, f_j is the fitness of type j, u_{ji} is the mutation rate from type j to type i and φ is the average fitness of the population given by $\varphi = \sum_j f_j x_j$. In our case the mutation matrix is given quite easily. There is only one allowed mutation, that is from type 0 to type 1 with a rate u. Hence, $u_{01} = u$ and $u_{10} = 0$. The mutation matrix U is then given by,

$$U = \begin{pmatrix} 1-u & u \\ 0 & 1 \end{pmatrix} \tag{2}$$

Since our population consists of only two types, we can write the quasi species equation in terms of the frequency of type 0 alone. Let x_0 be the frequency of type 0 in the population. Then, the frequency of type 1 is given by $x_1 = 1 - x_0$. The quasi species equation can then be written as,

$$\begin{aligned} \frac{\mathrm{d}x_0}{\mathrm{d}t} &= f_0(1-u)x_0 + f_1 u(1-x_0) - \varphi x_0 \\ &= x_0[f_0(1-u) - \varphi] \end{aligned} \tag{3}$$

The equilibrium frequency of type 0 can be obtained by setting $\frac{\mathrm{d}x_0}{\mathrm{d}t} = 0$. This implies, $x_0^* = 0$ or $\varphi = f_0(1-u)$. The mixed state solution leads to ,

$$x_0^* = \frac{f_0(1-u) - f_1}{f_0 - f_1} \tag{4}$$

Instead of taking derivatives to do stability analysis to check which of the two solutions is stable, we can rewrite the quasi species equation as,

$$\frac{\mathrm{d}x_0}{\mathrm{d}t} = x_0(x_0 - x_0^*)(f_0 - f_1) \tag{5}$$

From the above equation, we can see that if $f_0 > f_1$, then the solution x_0^* is stable mixed equilibrium state.

1. For the first part of the problem, we have $f_0 = 1.001$ and $f_1 = 1$. For these parameters, the equilibrium frequencies are,

$$x_0^* = 0 \quad \text{and} \quad x_0^* = \frac{1.001 \times 0.99 - 1}{1.001 - 1} < 0 \tag{6}$$

Thus the only stable equilibrium state is $x_0^* = 0$. This means that the population will eventually be taken over by type 1. The frequency of type 0 will eventually go to zero.

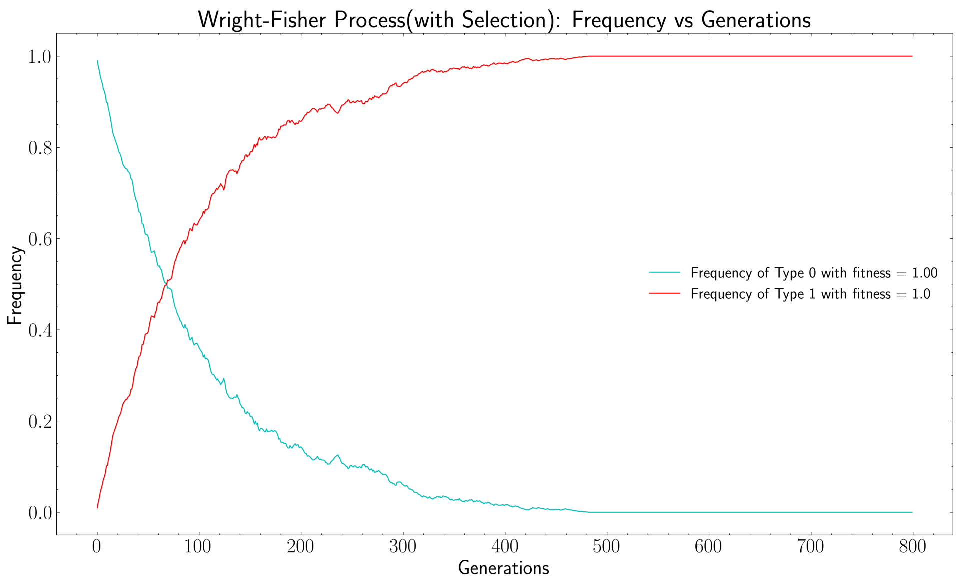

Figure 1: Wright Fisher process with selection and mutation: $f_0 = 1.001$ and $u = 0.01$

2. For the second part of the problem, we have $f_0 = 1.1$ and $f_1 = 1$. The initial frequencies are $x_0 = 0.5$ and $x_1 = 0.5$. For these parameters, the equilibrium frequencies are,

$$x_0^* = 0 \quad \text{and} \quad x_0^* = \frac{1.1 \times 0.99 - 1}{1.1 - 1} = 0.89 \tag{7}$$

Thus the stable equilibrium state is $x_0^* = 0.89$, where the stable solution is the mixed state. This means that the population will eventually reach an equilibrium state where the frequency of type 0 is 0.89.

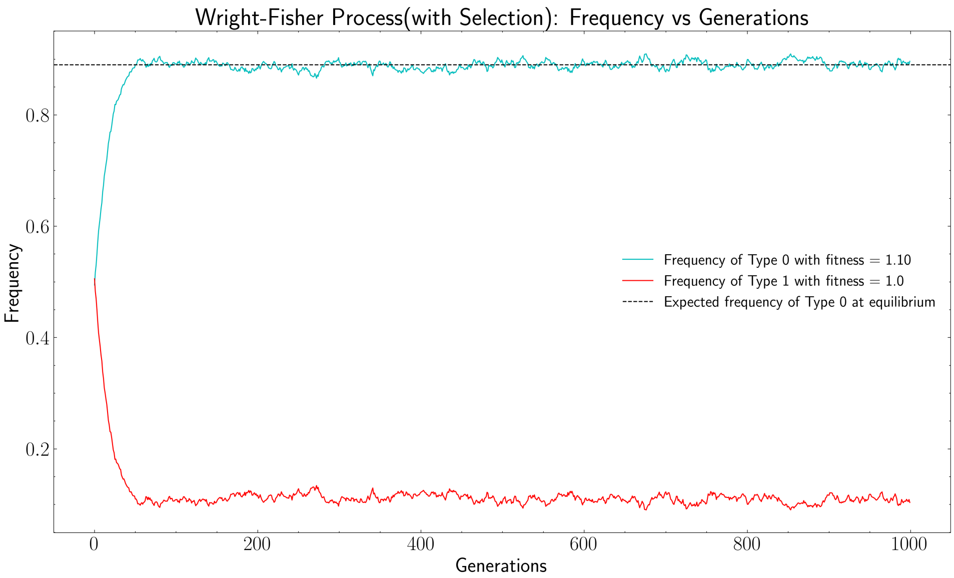

Figure 2: Wright Fisher process with selection and mutation: $f_0 = 1.1$ and $u = 0.01$

3. For the third part of the problem, we have $N = 100$. The equilibrium frequencies are the same as in the second part of the problem, that is $x_0^* = 0.89$. However, since the population size is smaller, the fluctuations around the equilibrium state will be larger.

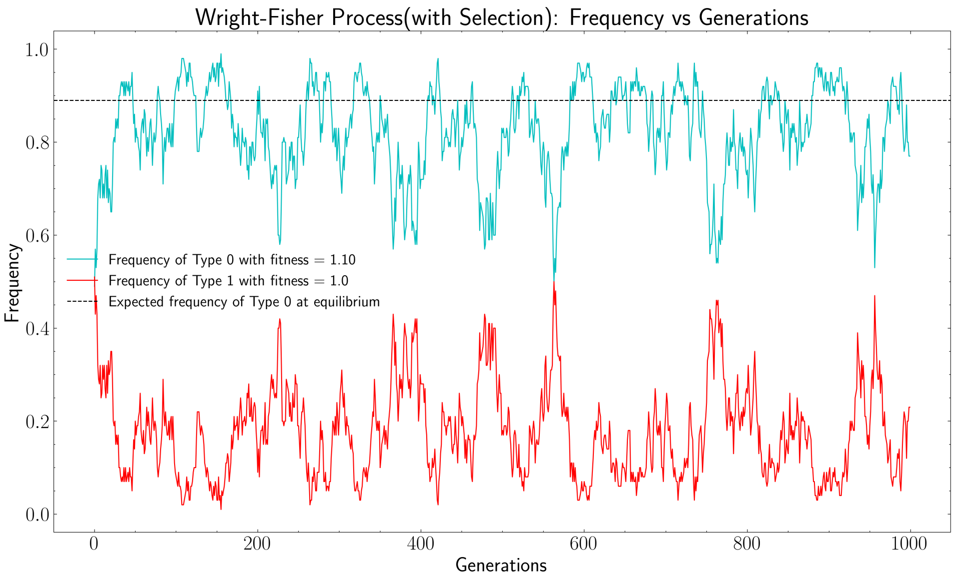

Figure 3: Wright Fisher process with selection and mutation: $f_0 = 1.1$, $u = 0.01$ and $N = 100$

A small note on algorithm implementation. To use mutation and selection together in the Wright-Fisher process, the selection process can be done without using a nested for loop. Since to probabilities of selection are bernoulli random variables, the sum of these together will be a binomial random variable. Hence, we can directly sample the number of offspring of each type from a binomial distribution.